

VECTOR SPACE & SUBSPACES

Complete Study Guide with Practice Questions

Mathematics — EC Engineering | SVNIT Surat

Fields · Vector Spaces · Subspaces · Linear Independence & Dependence · Coordinates · Bases & Dimension · Gram-Schmidt Orthonormalisation · Orthonormal Basis · Orthogonal Projection

SECTION 1 — FIELDS

1.1 Definition of a Field

A **Field F** is a set with two binary operations (+) and ($\times$) satisfying closure, associativity, commutativity, identity elements (0 and 1), additive & multiplicative inverses, and distributivity. Every element except 0 must have a multiplicative inverse.

Set	Field?	Reason
$\mathbb{R}$ (Real numbers)	YES	All axioms satisfied
$\mathbb{Q}$ (Rationals)	YES	Closed under all operations
$\mathbb{C}$ (Complex numbers)	YES	Includes inverses
$\mathbb{Z}$ (Integers)	NO	No multiplicative inverse (e.g. $1/2 \notin \mathbb{Z}$)
$\mathbb{N}$ (Natural numbers)	NO	No additive inverse, no 0

Practice Questions

Q1. Is the set of all rational numbers a field under usual + and $\times$? Justify.

Q2. Show that $\mathbb{Z}_5 = \{0,1,2,3,4\}$ with addition and multiplication modulo 5 is a field.

Q3. Explain why the set of integers is NOT a field.

SECTION 2 — VECTOR SPACES

2.1 Definition

A **Vector Space** V over a field F is a non-empty set with vector addition and scalar multiplication satisfying 10 axioms. Elements of V are called vectors; elements of F are called scalars.

2.2 The 10 Axioms

#	Axiom	Name
1	$u + v \in V$	Closure under addition
2	$u + v = v + u$	Commutativity
3	$(u+v)+w = u+(v+w)$	Associativity
4	$\exists 0 \in V : v + 0 = v$	Additive Identity
5	$\exists -v : v + (-v) = 0$	Additive Inverse
6	$c \cdot v \in V$	Closure under scalar mult.
7	$c(u+v) = cu + cv$	Distributive (over vectors)
8	$(c+d)v = cv + dv$	Distributive (over scalars)
9	$c(dv) = (cd)v$	Scalar assoc.
10	$1 \cdot v = v$	Scalar identity

2.3 Standard Examples of Vector Spaces

Space	Elements	Field	Dimension
$\mathbb{R}^n$	n-tuples $(x_1, \dots, x_n)$	$\mathbb{R}$	n
P_n	Polynomials degree $\leq n$	$\mathbb{R}$	$n+1$
$M_{m \times n}$	All $m \times n$ matrices	$\mathbb{R}$	$m \cdot n$
$C[a,b]$	Continuous functions on $[a,b]$	$\mathbb{R}$	Infinite
$\mathbb{C}^n$	n-tuples of complex numbers	$\mathbb{C}$	n

♦ Zero vector 0 is always unique in a vector space.

♦ $c \cdot 0 = 0$ and $0 \cdot v = 0$ for all $c \in F, v \in V$.

Practice Questions

Q1. Show that the set of all 2×2 matrices over $\mathbb{R}$ forms a vector space under matrix addition and scalar multiplication.

Q2. Prove that the set of all polynomials of degree ≤ 2 is a vector space over $\mathbb{R}$.

Q3. Is the set $V = \{(x,y) : x,y \geq 0\}$ a vector space over $\mathbb{R}$? Justify.

Q4. Verify that $C[0,1]$ (continuous functions on $[0,1]$) is a vector space.

Q5. Show that the set of all solutions of the homogeneous system $AX = 0$ forms a vector space.

SECTION 3 — SUBSPACES

3.1 Definition and Subspace Test

A non-empty subset W of a vector space V is a **subspace** if it is itself a vector space under the same operations. Instead of checking all 10 axioms, only 3 conditions are needed:

SUBSPACE TEST (3 conditions):

1. $0 \in W$ (zero vector in W)

2. $u, v \in W \Rightarrow u + v \in W$ (closed under addition)

3. $u \in W, c \in F \Rightarrow c \cdot u \in W$ (closed under scalar multiplication)

Compact form: $u, v \in W \Rightarrow au + bv \in W$ for all $a, b \in F$

3.2 Examples

Subset W	Space V	Subspace?	Reason
$\{(x, y, z) : x+y+z=0\}$	$\mathbb{R}^3$	YES	Plane through origin
$\{(x, y, z) : x+y+z=1\}$	$\mathbb{R}^3$	NO	0 not in W
$\{(x, y) : x \geq 0\}$	$\mathbb{R}^2$	NO	Not closed under scalar mult.
Symmetric $n \times n$ matrices	$M_{n \times n}$	YES	Sum & scalar of sym. is sym.
Polynomials with even degree	P_n	YES	Closed under + and scalar

♦ Every vector space has at least two subspaces: $\{0\}$ (trivial) and V itself.

♦ Intersection of two subspaces is always a subspace.

Practice Questions

Q1. Show that $W = \{(x, y, z) \in \mathbb{R}^3 : 2x - y + 3z = 0\}$ is a subspace of $\mathbb{R}^3$.

Q2. Is $W = \{(x, y) : xy = 0\}$ a subspace of $\mathbb{R}^2$? Justify.

Q3. Prove that the intersection of two subspaces W_1 and W_2 of V is a subspace.

Q4. Show that $W = \{A \in M_{n \times n} : A \text{ is symmetric}\}$ is a subspace of $M_{n \times n}$.

Q5. Determine if $W = \{p(x) \in P_n : p(0) = 0\}$ is a subspace of P_n .

Q6. Is the union of two subspaces always a subspace? Give example or counterexample.

SECTION 4 — LINEAR INDEPENDENCE & DEPENDENCE

4.1 Definitions

Consider vectors $v_1, v_2, \dots, v_n$ and scalars $c_1, c_2, \dots, c_n \in F$:

$$c_1 \cdot v_1 + c_2 \cdot v_2 + \dots + c_n \cdot v_n = 0$$

L.I. (Linearly Independent): only solution is $c_1 = c_2 = \dots = c_n = 0$

L.D. (Linearly Dependent): at least one $c_i \neq 0$ satisfies the equation

4.2 Key Properties

Property	L. Independent	L. Dependent
Trivial solution	Only solution	Not the only solution
One vector as combo of others	NOT possible	Always possible
Includes zero vector?	Never	Always (set with 0 is L.D.)
Single vector $\{v\}$?	If $v \neq 0$	If $v = 0$
Rank of matrix formed	Rank = n	Rank < n

4.3 Method to Check (Using Matrix)

Step 1: Form matrix A with the given vectors as rows (or columns)

Step 2: Apply Row Reduction (Echelon Form)

Step 3: Count non-zero rows = Rank r

If $r = n$ (no. of vectors) $\rightarrow$ Linearly INDEPENDENT

If $r < n \rightarrow$ Linearly DEPENDENT

♦ In $\mathbb{R}^n$, any set of more than n vectors is always linearly dependent.

♦ A set with the zero vector is always linearly dependent.

Practice Questions

Q1. Test if $v_1=(1,2,3)$, $v_2=(4,5,6)$, $v_3=(7,8,9)$ are L.I. or L.D.

Q2. Show that $\{1, x, x^2\}$ is linearly independent in P_2 .

Q3. Find the value of k for which $v_1=(1,2,3)$, $v_2=(1,1,1)$, $v_3=(1,k,k^2)$ are L.D.

Q4. Are the vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, $(1,1,1)$ linearly independent in $\mathbb{R}^3$?

Q5. Determine if the polynomials $p_1=1+x$, $p_2=1-x$, $p_3=1+x^2$ are L.I. in P_2 .

Q6. Show that any set of vectors containing the zero vector is linearly dependent.

SECTION 5 — COORDINATES, BASIS & DIMENSION

5.1 Span

The **span** of a set $S = \{v_1, v_2, \dots, v_n\}$ is the set of all linear combinations:

$$\text{span}(S) = \{c_1v_1 + c_2v_2 + \dots + c_nv_n : c_i \in F\}$$

span(S) is always a subspace of V.

5.2 Basis

A **Basis B** of V is a set that is: (1) Linearly Independent AND (2) Spans V. Every vector in V has a **unique** representation as a linear combination of basis vectors.

BASIS = L.I. + Spanning Set

Standard Bases:

$$\blacksquare^3: \{(1,0,0), (0,1,0), (0,0,1)\} \rightarrow e_1, e_2, e_3$$

$$P_2: \{1, x, x^2\}$$

$$M_{2 \times 2}: \{[1,0;0,0], [0,1;0,0], [0,0;1,0], [0,0;0,1]\}$$

5.3 Coordinates

If $B = \{b_1, \dots, b_n\}$ is a basis of V, every $v \in V$ has a unique representation $v = c_1b_1 + \dots + c_nb_n$. The coordinate vector of v w.r.t. B is:

$$[v]_B = (c_1, c_2, \dots, c_n)^T$$

Example: $B = \{(1,1), (1,-1)\}$ in $\blacksquare^2$, $v = (3,1)$

Solve: $c_1(1,1) + c_2(1,-1) = (3,1) \rightarrow c_1=2, c_2=1$

So $[v]_B = (2, 1)^T$

5.4 Dimension

The **dimension** of V is the number of vectors in any basis of V. All bases of a finite-dimensional space have the same number of elements.

Vector Space	Standard Basis	Dimension
$\blacksquare^n$	$e_1, e_2, \dots, e_n$	n
P_n	$\{1, x, x^2, \dots, x^n\}$	$n+1$
$M_{m \times n}$	Matrices with single 1	$m \cdot n$
$M_{n \times n}$ symmetric	$e_{ij} + e_{ji} (i \leq j)$	$n(n+1)/2$
$\{0\}$	Empty set	0

♦ **Rank-Nullity Theorem:** $\text{rank}(A) + \text{nullity}(A) = n$ (no. of columns)

♦ If $\dim(V) = n$: any n L.I. vectors = basis; any $n+1$ vectors = L.D.

Practice Questions

Q1. Find a basis and dimension of the subspace $W = \{(a,b,c) : a-b+c=0\}$ of $\mathbb{R}^3$.

Q2. Find the coordinates of $v = (2,-1,3)$ w.r.t. the basis $B = \{(1,1,0), (0,1,1), (1,0,1)\}$.

Q3. Show that $\{1+x, x+x^2, 1+x^2\}$ is a basis of P_2 .

Q4. Find the dimension of the solution space of the system: $x+y+z=0$, $x-y+2z=0$.

Q5. If $W = \text{span}\{v_1, v_2, v_3\}$ where $v_1=(1,2,1)$, $v_2=(2,1,3)$, $v_3=(1,-1,2)$, find $\dim(W)$.

Q6. Extend the set $\{(1,1,0,0), (0,0,1,1)\}$ to a basis of $\mathbb{R}^4$.

Q7. State and prove the Rank-Nullity theorem.

SECTION 6 — GRAM-SCHMIDT ORTHONORMALISATION

6.1 Inner Product and Orthogonality

Inner Product in $\mathbb{R}^n$: $u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n$

Norm (length): $\|v\| = \sqrt{v \cdot v}$

Unit vector: $\hat{v} = v / \|v\|$

Orthogonal: $u \cdot v = 0$ (perpendicular)

Orthonormal: $u \cdot v = 0$ AND $\|u\| = \|v\| = 1$

6.2 Projection Formula

Projection of v onto u :

$$\text{proj}_u(v) = (v \cdot u / u \cdot u) \cdot u$$

Component of v perpendicular to u :

$$v - \text{proj}_u(v)$$

6.3 Gram-Schmidt Process (Step by Step)

Given any basis $\{v_1, v_2, \dots, v_n\}$, convert to orthonormal basis $\{e_1, e_2, \dots, e_n\}$:

STEP 1: $u_1 = v_1$

STEP 2: $u_2 = v_2 - \text{proj}_{u_1}(v_2)$

$$= v_2 - (v_2 \cdot u_1 / u_1 \cdot u_1) u_1$$

STEP 3: $u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3)$

$$= v_3 - (v_3 \cdot u_1 / u_1 \cdot u_1) u_1 - (v_3 \cdot u_2 / u_2 \cdot u_2) u_2$$

STEP k: $u_k = v_k - \sum_{j=1}^{k-1} [(v_k \cdot u_j / u_j \cdot u_j) \cdot u_j]$

NORMALIZE: $e_k = u_k / \|u_k\|$ for each k

MNEMONIC: 'Subtract ALL previous projections, then normalize'

6.4 Worked Example

Given: $v_1=(1,1,0)$, $v_2=(1,0,1)$, $v_3=(0,1,1)$

$$u_1 = (1,1,0) \quad \|u_1\|^2 = 2$$

$$\begin{aligned} u_2 &= (1,0,1) - [(1 \cdot 1 + 0 \cdot 1 + 1 \cdot 0)/2](1,1,0) \\ &= (1,0,1) - (1/2)(1,1,0) = (1/2, -1/2, 1) \quad \|u_2\|^2 = 3/2 \end{aligned}$$

$$\begin{aligned} u_3 &= (0,1,1) - [(0+1+0)/2](1,1,0) - [(0-1/2+1)/(3/2)](1/2, -1/2, 1) \\ &= (0,1,1) - (1/2)(1,1,0) - (1/3)(1/2, -1/2, 1) \\ &= (0,1,1) - (1/2, 1/2, 0) - (1/6, -1/6, 1/3) = (-2/3, 2/3, 2/3) \end{aligned}$$

$$\begin{aligned} e_1 &= u_1/\|u_1\| = (1/\sqrt{2}, 1/\sqrt{2}, 0) \\ e_2 &= u_2/\|u_2\| = (1/\sqrt{6}, -1/\sqrt{6}, 2/\sqrt{6}) \\ e_3 &= u_3/\|u_3\| = (-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}) \end{aligned}$$

Practice Questions

Q1. Apply Gram-Schmidt to $\{(1,0,0), (1,1,0), (1,1,1)\}$ in $\mathbb{R}^3$.

Q2. Find an orthonormal basis for the subspace spanned by $(1,1,1)$ and $(1,2,3)$.

Q3. Apply Gram-Schmidt to $\{1, x, x^2\}$ in P^2 with inner product $= \int_{-1}^1 p(x)q(x)dx$.

Q4. Find an orthonormal basis for $W = \text{span}\{(1,1,0,0), (0,1,1,0), (0,0,1,1)\}$ in $\mathbb{R}^4$.

Q5. Apply Gram-Schmidt to $\{(1,2,0), (0,1,1), (1,1,1)\}$ to get an orthonormal basis.

SECTION 7 — ORTHONORMAL BASIS & ORTHOGONAL PROJECTION

7.1 Orthonormal Basis

A basis $\{e_1, e_2, \dots, e_n\}$ is **orthonormal** if all vectors are unit vectors and mutually perpendicular:

$$e_i \cdot e_j = \delta_{ij} = \begin{cases} 1 & \text{if } i=j \text{ (unit vector)} \\ 0 & \text{if } i \neq j \text{ (perpendicular)} \end{cases}$$

Key Properties:

1. Coordinates are easy: $c_i = v \cdot e_i$
2. Matrix Q with orthonormal columns: $Q^T = Q^{-1}$ (Orthogonal matrix)
3. $\|v\|^2 = c_1^2 + c_2^2 + \dots + c_n^2$ (Parseval's identity)

7.2 Orthogonal Matrix

Matrix Q is orthogonal if: $Q^T Q = Q Q^T = I$

Equivalently: $Q^{-1} = Q^T$

Property: $\det(Q) = \pm 1$

Example: Rotation matrix $[\cos\theta, -\sin\theta; \sin\theta, \cos\theta]$ is orthogonal

7.3 Orthogonal Projection

The **orthogonal projection** of v onto a subspace W gives the closest point in W to v . The error $(v - \text{proj})$ is perpendicular to W .

Projection onto a single vector u :

$$\text{proj}_u(v) = (v \cdot u / u \cdot u) \cdot u$$

Projection onto subspace W with orthonormal basis $\{e_1, \dots, e_k\}$:

$$\text{proj}_W(v) = (v \cdot e_1)e_1 + (v \cdot e_2)e_2 + \dots + (v \cdot e_k)e_k$$

Projection Matrix P : $P = A(A^T A)^{-1} A^T$

where A has basis vectors of W as columns

Properties: $P^2 = P$ (idempotent), $P^T = P$ (symmetric)

7.4 Best Approximation Theorem

The projection $\text{proj}_W(v)$ is the vector in W closest to v , i.e., it minimizes $\|v - w\|$ for all $w \in W$. This is the basis for **least squares approximation**.

◆ Applications: Least squares, Fourier series, Signal processing, QR decomposition.

Practice Questions

Q1. Find the projection of $v = (3, 4, 5)$ onto the subspace $W = \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

Q2. Find the projection matrix P that projects $\mathbb{R}^3$ onto the line spanned by $(1, 2, 2)$.

Q3. Verify that for the projection $p = \text{proj}_u(v)$, the vector $v - p$ is orthogonal to u .

Q4. Find the best approximation to $v=(1,2,3,4)$ in $W = \text{span}\{(1,1,0,0),(0,1,1,0)\}$.

Q5. Show that if Q is an orthogonal matrix, then $\|Qv\| = \|v\|$ for all v .

Q6. Find the QR decomposition of $A = [(1,1),(1,2),(1,3)]$ using Gram-Schmidt.

SECTION 8 — MASTER CHEAT SHEET

Concept	Definition	Key Test / Formula
Field F	Set with $+$ and $\times$ satisfying all axioms	Every non-zero element has mult. inverse
Vector Space V	Set over field F with 10 axioms	Check all 10 axioms
Subspace W	Subset that is also a vector space	$0 \in W$, closed under $+$ and scalar mult.
L. Independent	Only trivial solution to $c_1v_1 + \dots + c_nv_n = 0$	Rank = n
L. Dependent	Non-trivial solution exists	Rank < n
Span(S)	All linear combinations of S	Always a subspace
Basis B	L.I. + Spanning set	Unique representation for every v
Dimension	Number of basis vectors	Same for all bases of V
Orthogonal	$u \cdot v = 0$	Perpendicular vectors
Orthonormal	$e_i \cdot e_j = \delta_{ij}$	Unit + perpendicular
Projection	Closest point in subspace W to v	$\text{proj}_W(v) = (v \cdot u / u \cdot u)u$
Gram-Schmidt	Basis $\rightarrow$ Orthonormal basis	Subtract projections, normalize

Hierarchy

Field $F \supset$ Vector Space $V \supset$ Subspace W

Inside V : Span $\rightarrow$ Basis (L.I. + Spanning) $\rightarrow$ Dimension (size of basis)

Basis $\rightarrow$ [Gram-Schmidt] $\rightarrow$ Orthonormal Basis $\rightarrow$ [Projection] $\rightarrow$ Best Approximation

SECTION 9 — EXAM-STYLE QUESTIONS

Exam-Style Questions (2–5 marks)

Q1. Define a vector space and verify that $\mathbb{R}^3$ is a vector space over $\mathbb{R}$.

Q2. State the subspace criterion. Show that $W = \{(x, y, z) : x=y\}$ is a subspace of $\mathbb{R}^3$.

Q3. Define linear independence. Show that e_1, e_2, e_3 (standard basis) are L.I. in $\mathbb{R}^3$.

Q4. Find a basis for the null space (kernel) of $A = [(1,2,3), (4,5,6), (7,8,9)]$.

Q5. Show that $\dim(W_1 \cap W_2) \geq \dim(W_1) + \dim(W_2) - \dim(V)$ (Dimension formula).

Q6. Apply Gram-Schmidt to $\{(2,1), (3,2)\}$ in $\mathbb{R}^2$ to get an orthonormal basis.

Q7. Find the orthogonal projection of $(1,2,3)$ onto the plane $x + y + z = 0$.

Q8. If $\{v_1, v_2, \dots, v_n\}$ is a basis of V and $T: V \rightarrow W$ is a linear map, prove $\text{span}\{T(v_1), \dots, T(v_n)\} = \text{range}(T)$.

Q9. Show that every orthogonal set of non-zero vectors is linearly independent.

Q10. State Gram-Schmidt orthonormalization process and apply it to $\{(0,1,1),(1,0,1),(1,1,0)\}$ in $\mathbb{R}^3$.